Daan de Graaf

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Work

2023 – present **PhD Candidate on Programmability in Database Systems**, *Eindhoven University of Technology*, Eindhoven
My research focuses on efficient execution of user-defined graph algorithms in databases. My work appears at top-tier database conferences such as ACM SIGMOD, VLDB and CIDR. I built the query lowering pipeline for the AvantGraph graph query engine, and implemented native support for user-defined algorithms. As part of the SciLake (Horizon Europe) project I worked with researchers from different disciplines to help them write and execute their custom algorithms over massive graphs. I have lectured on the topic of query processing on modern hardware, and supervised student projects on graph storage, in-database sorting and query compilation. Supervisors: Nikolay Yakovets and George Fletcher. **Expected defence: June 2027.**

2019 – 2021 **Software Engineer**, *Google LLC*, London
I was part of the Display Ads Monetization team responsible for the pipeline calculating and disbursing billions of USD in advertising revenue to AdSense and YouTube partners every month. I led an effort to cut pipeline runtime six-fold to stay within its time budget. I personally designed and built a separate pipeline for high-value partners that processed well over a billion USD monthly. Next to my core responsibilities, I also taught introductory classes on Rust and contributed to the V8 JavaScript engine that powers Google Chrome.

2018 – 2019 **Software Engineer**, *Prodrive Technologies B.V.*, Eindhoven
After 2 months as an Embedded Systems Engineer, I was selected for the newly created Developer Infrastructure team, where I worked with teams across the company to improve the reliability of their products and increase developer productivity.

Notable Projects

2023 – present **The GraphAlg Language**, wildarch.dev/graphalg
GraphAlg is a domain-specific programming language for graph algorithms that is designed to be integrated into existing database systems. It is natively supported in the AvantGraph graph query engine, with additional backends available for the relational systems Umbra and DuckDB. GraphAlg takes existing theoretical works, such as the formal MATLANG language for matrix manipulation, and turns these into practical extensions of real database systems, with a focus on achieving high performance through aggressive query optimization. The GraphAlg compiler, as well as the online playground and tutorial, are open source software.

May–Jul 2026 **Iteration in Relational Database Systems**, *Technical University of Munich*, Munich
During a 3-month research visit at the chair for database systems I investigated how queries with loops can be represented, optimized and executed efficiently in a modern relational query engine. I designed a highly flexible loop construct for SQL that subsumes multiple existing proposals, yet is simple enough to be implemented with minimal engine changes. The prototype, implemented in the Umbra database system, outperformed competing relational, graph-native and Datalog systems on both graph analytics and Datalog workloads. This work was the basis for a CIDR publication, and is now available in standard Umbra releases.

Selected Publications

CIDR 2027 **Running Loops around Recursive SQL *(to appear)***
Daan de Graaf, Altan Birler, Nikolay Yakovets, Thomas Neumann
A loop operator for SQL to cleanly and efficiently express graph algorithms and Datalog programs.

SIGMOD 2027 **Algorithm Support for Graph Databases, Done Right *(to appear)***
Daan de Graaf, Robert Brijder, Soham Chakraborty, George Fletcher, Bram van de Wall, Nikolay Yakovets
Introduces GraphAlg, a domain-specific language for graph algorithms, designed to be integrated into database systems.

VLDB 2026 **GraphAlg Playground: An Online Platform for Learning and Experimenting with the GraphAlg Language**
Daan de Graaf, Robert Brijder, Soham Chakraborty, George Fletcher, Bram van de Wall, Nikolay Yakovets
Demonstration of an online playground and tutorial for the GraphAlg language.

OOPSLA 2023 **Randomized Testing of Byzantine Fault Tolerant Algorithms *(ACM SIGPLAN Distinguished Paper award)***
Levin N. Winter, Florena Buse, **Daan de Graaf**, Klaus von Gleissenthall, Burcu Kulahcioglu Ozkan
A simple yet effective method for finding bugs in Byzantine fault tolerant algorithms through fuzzing.

Education

2021 – 2023 **Computer Engineering MSc**, *Delft University of Technology*, Delft, Cum laude
Specializations in Compiler Construction, Data-Intensive Systems Engineering and Computer Architecture. Thesis: *Efficient Processing of User-Provided Graph Algorithms in a Graph Database*. The initial prototype of GraphAlg was developed during the thesis. GPA: 9.0/10

2015 – 2018 **Computer Science BSc**, *Eindhoven University of Technology*, Eindhoven, Cum laude
Electives in Security, Artificial Intelligence and Computer Architecture. GPA: 8.55/10.0

Competences

Expertise	Query processing, Compiler construction, Performance engineering, Large-scale software development
Technical	C/C++, LLVM/MLIR, Rust, SQL, Linux
Soft skills	Academic writing, Public speaking, Project management

Languages

Dutch: Native speaker **English:** Fluent **Chinese:** HSK1

Community Service

2025 **Student Volunteer**, *Conference on Innovative Data Systems Research (CIDR)*

2024 **Student Volunteer**, *IEEE International Conference on Data Engineering (ICDE)*

2024 – present **Reviewer**, *VLDB Journal*

2017 – 2018 **Chairman**, *Scala*, Eindhoven, studentencultuur.nl
Represented a collective of 9 cultural student associations with over 1000 members.

2016 – 2017 **President**, *AEGEE-Eindhoven*, Eindhoven, aegee.org
Represented the Eindhoven branch of the AEGEE student association both locally and internationally.

Interests

Cooking I enjoy trying out new recipes and learning the science behind cooking to make better food. In particular, I am interested in Asian cuisine and bread-making. My collection of recipes is available on GitHub.

Climbing I enjoy bouldering and sport climbing, both indoors and outdoors.